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Machine Learning in Robotic Ultrasound Imaging: Challenges and Perspectives

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Keywords

deep learning, segmentation, registration, ultrasound image analysis, ultrasound simulation, reinforcement learning, learning from demonstration, ultrasound physics, ethics and regulations, medical robotics

Abstract

This article reviews recent advances in intelligent robotic ultrasound imaging systems. We begin by presenting the commonly employed robotic mechanisms and control techniques in robotic ultrasound imaging, along with their clinical applications. Subsequently, we focus on the deployment of machine learning techniques in the development of robotic sonographers, emphasizing crucial developments aimed at enhancing the intelligence of these systems. The methods for achieving autonomous action reasoning are categorized into two sets of approaches: those relying on implicit environmental data interpretation and those using explicit interpretation. Throughout this exploration, we also discuss practical challenges, including those related to the scarcity of medical data, the need for a deeper understanding of the physical aspects involved, and effective data representation approaches. We conclude by highlighting the open problems in the field and analyzing different possible perspectives on how the community could move forward in this research area.

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1. INTRODUCTION

Over the past few decades, and particularly in the last 10 years, the advancement of autonomous medical robots has attracted increasing attention (1, 2). Recent advances in fundamental sensing systems and artificial intelligence have made progress toward the creation of intelligent robotic colleagues that can work with medical staffs in hospitals (3–5). Among the various subfields of medical robotics, robotic sonography has particularly garnered attention from both the scientific and industrial sectors (6–8). This surge in interest can be attributed to the fact that robotic ultrasound (US) examinations are generally noninvasive compared with procedures involving other surgical robots, resulting in fewer ethical, legal, and regulatory concerns.

Due to the advantages of being portable, real-time, and free of ionizing radiation, medical US has gained widespread use in primary healthcare. The rapid evolution of medical imaging modalities throughout the last century has transformed imaging into an indispensable element of standard screening and diagnostic protocols. However, the development of interactive and dynamic imaging for treatment guidance and intervention has lagged behind. This can be attributed, in part, to the intricate patient and procedure-specific demands that such imaging entails. Moreover, its considerable reliance on user proficiency and usability considerations has also contributed to its slower advancement (9). To improve reproducibility and ensure consistent diagnosis, Salcudean et al. (10) introduced the use of robotic systems to assist in US acquisition. By accurately maneuvering the US probe, robotic US systems (RUSSs) are expected to standardize examination protocols and optimize imaging quality (11, 12). In light of the increasing demand for healthcare interventions and the uneven distribution of experienced sonographers, there is a pressing need for the development of RUSSs with a high degree of autonomy. The applications for representative clinical scenarios are summarized in **Figure 1**.

Robotic systems have demonstrated the potential to surpass human capabilities in certain domains. Advanced mechanical and material designs, coupled with innovative control methods, enable robots to operate remotely in challenging environments, such as the lunar surface (26) and the deep sea (27). Nevertheless, there is still a big gap between robotic systems and humans regarding their capabilities for deep understanding and modeling of the dynamic world. Similar to their human counterparts, intelligent robotic sonographers should possess the ability to comprehend human anatomy, diseases, physiology, and the physics of US imaging. Only by possessing these capabilities can a robotic sonographer reason about proper actions while performing US examinations and, at the same time, taking the utmost consideration for patient safety. The two key abilities in this context are perception and execution, which are tightly entangled and can mutually boost each other. Precise scene understanding will benefit the action reasoning process, while accurate maneuvering of the US probe will further facilitate understanding. Unfortunately, such intelligent systems have yet to emerge within the community. But thanks to the boom of machine learning and deep learning, researchers are approaching this ultimate goal from various perspectives. Jiang et al. (28) introduced the concept of "the language of sonography" (p. 1) by inferring the underlying reward function from a few US examination demonstrations from experts to understand the intention of the sonographer. Baumgartner et al. (29) proposed SonoNet, a deep convolutional neural network (CNN), to automatically detect 13 fetal standard planes in real time. In addition, when searching for standard planes, Droste et al. (30) presented a behavior-cloning framework to mimic sonographers.

Several survey articles have investigated RUSSs from different perspectives. For instance, Jiang et al. (6) discussed advancements in terms of the complexity of the technologies, while others have categorized papers based on autonomy levels (7, 8). In contrast, this survey article uniquely focuses on the technologies aimed at enhancing the intelligence of RUSSs. Section 2 presents an

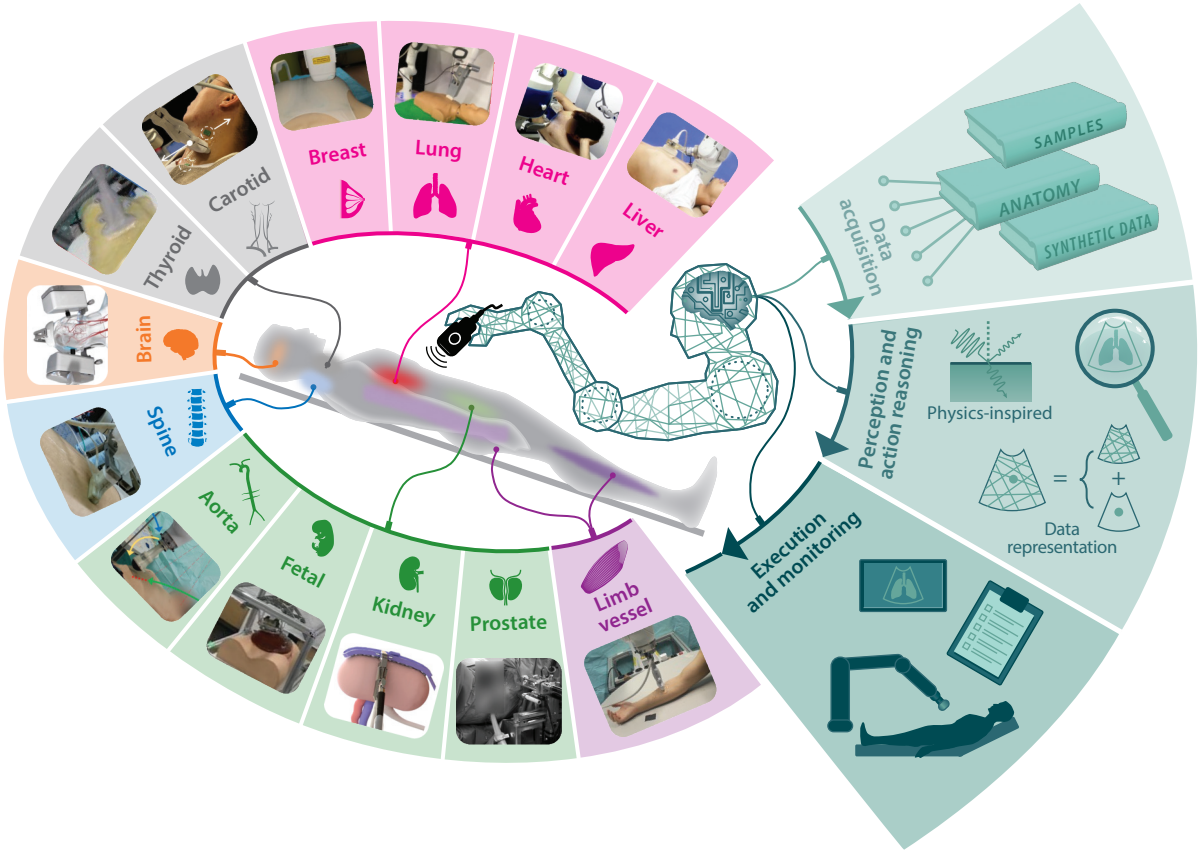


Figure 1

Robotic ultrasound imaging systems. (*Left*) Typical clinical applications in robotic ultrasound imaging. Starting at the top right and proceeding counterclockwise: liver (13), heart (14), lung (15), breast (16), carotid (17), thyroid (18), brain (19), spine (20), aorta (21), fetal (22), kidney (23), prostate (24), and limb vessel (25). (*Right*) Key components of an intelligent robotic ultrasound system. To autonomously maneuver an ultrasound probe toward diagnostic views, an intelligent robotic sonographer is envisioned to be capable of leveraging prior anatomical knowledge and analyzing real-time observations for action reasoning.

overview of the hardware and control algorithms in RUSSs. We then highlight studies using machine learning technologies to improve the understanding of complex natural and physiological scenes or to directly reason about proper actions based on implicit scene comprehension in Section 3. Additionally, we have examined potential solutions to tackle practical challenges, including the shortage of medical data, the necessity for considering physical aspects, and the implementation of effective data representation methods in Section 4. Finally, we conclude by presenting several perspectives for the future of intelligent RUSSs.

2. HARDWARE AND CONTROL IN ROBOTIC ULTRASOUND

To acquire high-quality US images for diagnostic or analytical purposes, RUSSs should be able to maneuver the US probe toward the target region while precisely regulating several acquisition parameters (11). These parameters primarily involve contact forces, positions, and orientations. The desired values are either determined from the clinician's side within the teleoperated/semiautonomous paradigms (10, 31) or derived from intuitive/intelligent algorithms

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(12, 32) running in the background. Consequently, the evolution of RUSSs' mechanism design and control strategy is prominently centered around enhancements in imaging quality, control accuracy, and execution safety. To highlight the main focus of applying machine learning to RUSSs, this section provides a brief summary of relevant hardware and control methods used for developing RUSSs. A more systematic summary can be found in a previous survey paper (6).

2.1. Mechanism Design

Safety stands as the highest priority in the development of RUSSs. The implementation of specially designed mechanisms can passively limit the applied forces on the patient and avoid potential failure of the sensor or the electrical system, thereby bolstering safety throughout US scanning. To this end, Tsumura & Iwata (33) developed a hardware platform to position a US probe for obstetric examination. The platform includes a linear actuator to limit the contact force and a ring-guide mechanism to passively adjust the US probe's posture relative to the body surface. To ensure safety at the actuator level, Wang et al. (34) investigated a customized spring-loaded ball clutch joint to passively restrict the exerted force.

To facilitate quick deployment and fast adaptation to different scanning tasks, it is also common to use general robotic manipulators to develop RUSSs (13, 35). In this case, the mechanism design focuses more on refining the US probe holder component. Tan et al. (16) proposed a dual arm system with a flexible spring-based US probe clamping device for breast US imaging. To passively adapt to the irregular contact contours, Facundo-Flores et al. (36) adopted a soft probe, which provides good surface adaptation and force transfer, as part of the RUSS. Wang et al. (37) developed a compliant joint-based adjusting mechanism attached to the robot manipulator to facilitate spine US examinations. Recently, a motorized probe holder for a robot manipulator has been reported. Bao et al. (38) designed a motor-spring-based mechanical US probe holder to enable online force adjustment and operation continuity. This design pattern can be a promising way to achieve compliant contact for safety and active posture alignment between the probe and contact surface for better image quality.

2.2. Control Method

In the context of US scanning, the full control of RUSSs can be decoupled into two subspaces: one-degree-of-freedom force control along the imaging direction of the probe and five-degree-of-freedom motion control for navigation along a scanning path (6). To orient a US probe, velocity-based hybrid force–motion control schemes are commonly used in the community (39, 40). Given that US examinations require extensive contact with patients, impedance/admittance control is frequently employed to dynamically modulate contact force and position, primarily for safety concerns (31, 41, 42). To suppress the accumulated control error, Wang et al. (43) proposed a practical full-task-space compliant control framework with motion estimation and feedback reinitialization for RUSSs. Nevertheless, the performance of impedance/admittance control relies heavily on the precise estimation of the environmental parameters, specifically stiffness and damping terms.

Due to the interpatient variation of tissue properties, varying levels of force are often required to obtain high-quality images (21, 44). To accommodate changes in contact conditions, Duan et al. (45) proposed a quadratic-programming-based control framework with variable impedance for US-guided scoliosis assessment. Xiao et al. (46) employed variable impedance control to regulate the contact force for better visualization in a human-in-the-loop environment. To effectively govern the contact dynamics, Santos & Cortesão (47) proposed a hierarchical control structure where the Cartesian force control takes the primary priority and orientation is controlled within

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the null space. Their work incorporated online estimation of contact stiffness, enabling control adaptation before contact. Alternatively, Kim et al. (48) utilized the Hunt–Crossley model to formulate a control law with online parameter estimation. Abbas et al. (49) devised an adaptive event-triggered force–motion control scheme for transversal abdomen scans. The control scheme combines an adaptive backstepping motion controller with a proportional–integral–derivative (PID) force controller, and the update of control input is triggered based on Lyapunov-based analysis. To integrate RUSSs into real-world scenarios, various efforts have been made to optimize the acquisition process using multisensor feedback. These optimizations include adjusting the US probe's posture (11, 12, 15, 50, 51), adapting contact forces (46, 52), refining scanning trajectories (39, 53), and so on.

The integration of robotic systems has showcased the capacity to elevate the precision and consistency of US image acquisition. We consider the soft mechanics and control techniques in RUSSs to be relatively well developed, and the key challenge lies in comprehending dynamic environments and action reasoning—a crucial hurdle on the path to realizing intelligent robotic sonographers. In the following sections, we provide some insights into the application of machine learning algorithms that try to address this challenge.

3. MACHINE LEARNING IN ROBOTIC ULTRASOUND

This section categorizes the machine learning approaches in RUSSs into modular and direct approaches for action reasoning (**Figure 2**). Direct approaches use advanced learning frameworks that continuously update and optimize for the best control strategy by directly observing the environment and output corresponding actions. Modular approaches, in contrast, include mainly control algorithms that integrate the outcomes of deep learning models into rule-based control laws. In modular approaches, intermediate perception results are explicitly extracted, while in direct approaches, the intermediate clues are implicitly processed inside a decision-making network. Both approaches have their own advantages and disadvantages. Direct approaches require less supervision from experts, while modular approaches are more predictable.

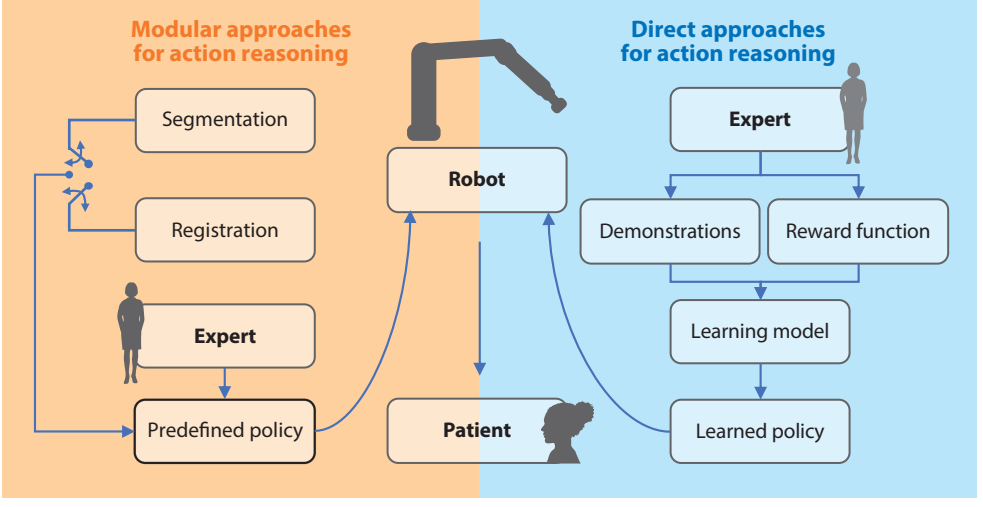


Figure 2

Modular and direct approaches for action reasoning. In modular approaches, the control strategy is defined by a human expert based on different intermediate results provided by machine learning models. Direct approaches use machine learning algorithms, where the action policy is directly learned from demonstrations or interactions.

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3.1. Modular Approaches for Action Reasoning

For human sonographers, action reasoning is based on their understanding of US image formation in US examinations. In modular approaches, machine learning algorithms explicitly model these implicit interpretation processes of sonographers. Two of the most commonly used analytical techniques are semantic segmentation and registration. The former provides morphological and positional information on the target anatomy, while the latter makes it feasible to leverage preoperative or generic models. This section discusses both techniques in detail.

3.1.1. Segmentation. Semantic segmentation of the US image is a commonly used method in RUSSs to specify and localize the target anatomy. In practice, the semantic segmentation process is also frequently carried out implicitly in the sonographer's mind during US examinations. To enable autonomous tracking of the target tissue, the outcomes of the segmentation process are incorporated into the robot's control loop. Such frameworks have been widely implemented for different US scanning tasks. Merouche et al. (54) implemented a fast-marching segmentation method to locate the vessel lumen, subsequently directing the robot to maintain the vessel at the center of the US view. Similarly, Akbari et al. (55) segmented seroma in breast US using a threshold method to provide feedback to the robot control loop.

However, such conventional segmentation methods often lack the intuitive understanding of the anatomical information because they rely on gray-level probability density functions. Therefore, the performance of these methods often suffers from noise, speckle, and artifacts. Recently, deep CNNs have emerged as a powerful alternative to conventional segmentation methods, demonstrating phenomenal success in medical image analysis (56). In general, there are two characteristics that need to be properly considered: contour and texture. Contour refers to the statistical shape model of the target anatomy, providing a robust cue for the model during segmentation tasks. However, not all anatomies exhibit stable statistical shapes, especially in cases involving lesions or abnormalities. Hence, the model should also be able to distinguish the segmentation targets based on their unique textures. In the context of CNNs, anatomy contour extraction depends largely on the network's capability to address large-scale relationships within the image, whereas the efficiency of texture recognition is closely linked to the network's performance in perceiving local details of images.

Currently, one of the most popular and effective network architectures for biomedical image segmentation is U-Net (57). Its success is attributed mainly to the skip connections at different scales of the encoder–decoder architectures. Such implementation ensures that the network can take advantage of fine details provided by high-level features and global information extracted from low-level features. Based on a real-time segmentation of tubular structures using U-Net, Jiang et al. (58) proposed a framework to simultaneously compute the radius of the vessel and optimize the vascular centerline to determine the scanning path. Koskinopoulou et al. (59) designed a dual robot system for venous intervention guided by a segmentation of the veins using Mask R-CNN, which is a segmentation network designed for simultaneous detection and segmentation of objects. The aforementioned two RUSSs have been validated on phantoms. To further validate the practical applicability of autonomous RUSSs in real-world conditions, Chen et al. (60) integrated an external camera with US vessel detection outcomes from U-Net. This enabled automatic US scanning and subsequent reconstruction of the patient's femoral artery. In an autonomous venous puncture mechanism designed by Chen et al. (61), a semisupervised segmentation network, semi-ResNeXt-Unet, was used to address the problem of insufficient labels. The accuracy of the vein segmentation was tested on volunteers and later used to guide the puncture robot. To improve the stability of artery segmentation in US images, Jiang et al. (62) introduced DopUS-Net, which combined Doppler images with B-mode images. During the autonomous scanning of radial

arteries, the presence of the Doppler effect is constantly monitored, and an out-of-plane fan motion of the US probe is executed when necessary to ensure good segmentation. Doppler images were also employed by Ning et al. (63) as weak supervision for the segmentation network.

However, it is worth noting that U-Net and its variants are not specifically designed for US images. The interpatient and even intermachine differences of US images are relatively larger than those observed in optical or CT images. As a result, distinct US-oriented learning approaches are necessary to pave the way toward stable and reliable segmentation performance. In this regard, recent works that have explicitly integrated US physics into learning models, such as CACTUSS (Common Anatomical CT-US Space) (64) and LOTUS (Learning to Optimize Task-Based US Representations) (65), appear to offer a promising solution, which is discussed in detail in Section 4.2.

3.1.2. Registration. Registration is a commonly employed way for RUSSs to map an unseen patient-specific scenario to a generic template, preoperative CT, or MRI. This is, in fact, what human sonographers do regularly: They use their full understanding of human anatomy acquired throughout their education and training to explore the anatomy of interest and navigate to required US views for a given clinical task. For robotic sonographers, CT or MRI atlases could play the role of learned mental mapping.

It should be noted that this section focuses only on the application of the registration approach on RUSSs, rather than providing a comprehensive summary of technical details about registration (which can be found in other articles, e.g., Reference 66). Registration techniques have often been used for path planning in developing autonomous RUSSs, particularly for the ones from preoperative CT or MRI to the current setup. To this end, Hennersperger et al. (67) employed a classic iterative-closest-point algorithm to align the patient surface extracted from a live RGB-D camera to preoperative MRI. Langsch et al. (68) performed a rigid registration between US volume and MRI/CT volume using the linear correlation of linear combination (LC^2) similarity metric to determine the scanning path for aorta. To tackle the interpatient variation, Virga et al. (21) applied nonrigid registration to elastically register the patient to a generic MRI-based template using a Gaussian mixture model. Specifically for articulated joint motion, Jiang et al. (25) proposed a nonrigid registration method to autonomously generate a scanning path for the limb examination based on a generic MRI template. The robustness and effectiveness of this approach have been evaluated on different volunteers.

The aforementioned approaches were proven to be robust in their scenarios, such as the aorta and limb artery examinations. However, for specific applications, such as thoracic examination or intervention within limited acoustic spaces, additional physical constraints should be further taken into consideration to develop an effective and robust system. Due to the high acoustic impedance of bone, an acoustic shadow will be cast and further hinder the visibility of the tissue underneath. To address this challenge, Jiang et al. (69) first proposed a skeleton graph–based nonrigid registration method enabling the transfer of a preplanned path in the limited intercostal space from CT template to patients. To further address the cumbersome tuning of hyperparameters, an evolved solution using a directed dense skeleton graph–based registration was proposed (70).

Besides path planning, registration has also been seen as a potential and promising solution to address the complex issue of patient movement during the scanning process. To develop a motion-aware robotic sonographer, Jiang et al. (71) presented a framework to autonomously detect a patient's motion and update the trajectory based on the registration between the markers' locations (sparse point cloud) obtained before and after the motion using an external camera. Additionally, an advanced study using the dense point clouds extracted from an RGB-D camera was proposed and validated on a human-like arm phantom with an uneven surface outliner (72).

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The aforementioned approaches were mainly developed based on extracted point clouds. Such approaches are good for robotic applications because they balance registration accuracy and computational efficiency; however, they usually cannot provide pixel-wise precise registration. To address this problem, Hennersperger et al. (67) carried out an online US–MRI deformable registration by computing the LC^2 similarity to ensure accurate path transfer. LC^2 was proposed for multimodal registration, and it has been reported that it can provide robust and reliable descriptions of different modalities (73). The modality-independent neighborhood descriptor (MIND) is another prominent metric developed for the same purpose (74). Most recently, Ronchetti et al. (75) introduced a generic framework to create expressive cross-modal descriptors using a CNN. The results demonstrated that the proposed method can outperform the previous similarity method in terms of computational efficiency and generalization capability across anatomies and imaging modalities.

3.2. Direct Approach for Action Reasoning

In this section, we discuss the recent advancements in robotic learning in RUSSs. In contrast to the last section, the focus here is on works that deduce appropriate actions based on the interaction through methods such as reinforcement learning (RL) and learning from demonstration (LfD).

3.2.1. Learning from interactions. RL is a famous interactive learning paradigm that has achieved superhuman performance in various games (76). The core idea of RL is to build a mapping between the observed states and the optimized actions through numerous interactions with the unknown environment, guided by predefined reward functions. This process can be modeled as a Markov decision process. Nowadays, with the development of deep learning, neural networks are utilized as powerful approximators when facing a complex state–action space. In the field of robotic US, initial attempts have been made to exploit the usage of RL. Unlike most RL tasks, where the true state of the agent is directly observable, most US acquisition tasks can only be modeled as partially observable Markov decision processes. Thus, the success of the search is determined largely by whether the model can correctly infer the true state (i.e., the relative position between the probe and the scanning target) from its observations (i.e., US images).

Hase et al. (77) implemented a deep Q-learning model to navigate the US probe toward the sacrum of the spine, while considering only two translational movements. Li et al. (78) extended the dimension of the action space of the Q-learning method from two to five to complete a similar acquisition task for spine US. They implemented a confidence map (79) as part of the reward function to include an image quality assessment in the optimization loop. Instead of using US images as the observation, Ning et al. (80) utilized a scenery image as the state representation and employed an RL-based proximal policy optimization approach to realize stable localization of the standard plane. However, the generalization ability of the aforementioned approaches is limited by the insufficient number of training data. To address this problem, Bi et al. (81) proposed to utilize a segmentation network to keep only the task-related features and wipe out the unnecessary background information from the US image. The training can then be performed with simulated artificial data rather than acquiring real patients' US volumes. Instead of using the segmented mask as an intermediate space to bridge between the simulation environment and real scenario, Li et al. (82) proposed to use simulated US images from CT as an intermediate domain for training of RL agent. During the inference stage, the real US images are transferred to the US simulation domain using a generative adversarial network (GAN) so that the training of RL can be done solely in the CT space.

3.2.2. Learning from demonstration. LfD is another approach to achieve autonomous US examinations, where the model learns directly from experts' demonstrations. LfD methods can

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be further categorized into guidance networks, where the model predicts actions based on US images, and assessment networks, which estimate preference scores for individual US images to reveal the intentions of sonographers during screening. As a valuable side benefit, the latter one can further be used to indicate "the language of sonography" (28).

To mimic how sonographers perform US examinations, Droste et al. (30) proposed US-GuideNet to predict the action of the operator as well as the goal position of the standard plane using imitation learning. The demonstrations consist of paired US frames and the orientations of the probe recorded from an inertial measurement unit sensor. A gated recurrent unit was implemented to exploit the information of sequential data. However, it remains a large challenge to extract practical clues from the US images for action prediction. As a result, Men et al. (83) integrated the gaze signal of the operator into the network to provide extra guidance. The proposed Multimodal-GuideNet is able to simultaneously estimate the gaze of the operator and the probe motion. Instead of only providing instructions to the operator during scanning, Deng et al. (84) implemented the learned policy directly from demonstrations to control the robot. The US frames together with contact forces and probe positions were recorded as expert demonstrations.

Rather than cloning the movement of the expert from the given demonstrations, another way of achieving LfD is through inverse reinforcement learning (IRL) (85), which infers a reward function from demonstrations. Regarding US standard plane acquisition, such a reward function refers to the preference of sonographers, demonstrating the anatomical and diagnostic value of the US image. The common assumption for IRL is the positive correlation between the reward value and the occurrence frequency of a given state in demonstrations (86). However, in most cases, the US navigation is not optimal, which indicates the existence of a large number of redundant actions in the demonstrations. To address the problem of suboptimal demonstrations, Burke et al. (87) proposed a probabilistic temporal ranking method. It assumed that the US image in the latter stage of the demonstration is more likely to have higher diagnostic importance than the image in an earlier stage. Such an assumption is realistic for US screening, where the search normally terminates at the point when the standard plane is properly visualized. By relating the importance of the US image to its relative position to the end point of the demonstration, Jiang et al. (28) proposed a global probabilistic spatial ranking. A feature disentanglement technique was also introduced to the network to improve its performance on unseen scenarios of ex vivo phantoms and in vivo carotid US data.

4. ADVANCEMENTS TOWARD INTELLIGENCE IN ROBOTIC ULTRASOUND

In this section, we demonstrate some important aspects that need to be properly addressed for RUSSs, i.e., data scarcity, the necessity of integrating physics into the design of deep learning networks, and efficient data representation.

4.1. Data Scarcity

For RUSSs to be successfully applied in clinical practice, they must be both robust and reliable. However, achieving these attributes presents several challenges. A primary obstacle is the limited availability of ground truth data essential for training machine learning algorithms. Traditional approaches to data collection not only consume significant time and resources but often necessitate expert involvement for accurate labeling. This creates bottlenecks in developing machine learning models. This section presents methodologies to either circumvent or minimize the requirement for labeled data.

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4.1.1. Ultrasound simulation. By employing physics-based US simulations, researchers can generate a wealth of synthetic but realistic data, serving as ground truth for machine learning models. There are three main approaches for US simulation, i.e., interpolative methods, physics-based generative models, and deep learning–based techniques. These methods aim to generate realistic 2D or 3D images by capturing the complexities of tissue interactions with US waves.

Interpolative methods, as implemented by d'Aulignac et al. (88) and Goksel & Salcudean (89), serve as a fast and efficient means of image reconstruction. These techniques employ previously acquired US images and use them as a basis to generate new images through interpolation algorithms. The approach excels in quick image processing and realistic domain representation. However, it struggles to accurately portray directional artifacts, and its capability to produce novel views is constrained by the set of recorded images.

When it comes to physics-based generative simulation, wave simulation serves as one of the most accurate, albeit computationally demanding, approaches. One of the most popular libraries, developed by Treeby & Cox (90), is the k-Wave toolbox, which relies on solving coupled first-order acoustic equations in the *k*-space pseudospectral time domain.

Following a different, computationally less expensive approach, convolution-based methods excel in visualizing subwavelength tissue structures by convolving a point spread function with a scatterer map (91). While effective for microscopic interactions, they lack accuracy in macroscopic correlations. Ray tracing, leveraging geometric optics principles, fills this gap by modeling US propagation as rays interacting with tissue. Combining convolution with ray tracing offers a comprehensive approach, capturing both microscopic and macroscopic interactions in US simulations (92).

Additionally, by leveraging presegmented medical imaging data, patient-specific US simulation can be acquired. For instance, Salehi et al. (93) used segmented MRI data as input and applied convolutional ray tracing for appearance optimization. Velikova et al. (64) used this approach for training a segmentation network only on simulated US data, based on CT and associated labels. An enhanced version of the ray-tracing simulation was presented by Mattausch et al. (94). Combining surface models with Monte Carlo surface interaction sampling for macroscopic interactions and convolution-based simulations for scattering phenomena enables the creation of an accurate representation of the underlying texture.

Deep learning–based techniques, particularly GANs, offer a data-driven mechanism for US image synthesis. Research such as that by Tom & Sheet (95) and Hu et al. (96) has exploited the capabilities of GANs to generate highly realistic US images. In addition, the application of CycleGANs, demonstrated by Vitale et al. (97), has shown potential in reconciling the discrepancies between simulated and real US images, making this a promising technique for training and validating machine learning models in robotic US applications.

4.1.2. Data augmentation. To increase the diversity of training data, another intuitive approach is data augmentation. Out-of-domain and synthetic images are created based on the training data using either conventional image transformation methods or deep learning–based generative models. As a traditional augmentation method, Zhang et al. (98) proposed a deep stacked transformation framework, which has been tested on different medical imaging modalities and demonstrated promising results in increasing generalization capability. Focusing on a kidney US segmentation task, Yin et al. (99) utilized a nonrigid image registration method to register and alter the shape of one kidney with other kidney samples in the training dataset so that the anatomical correctness of the augmented kidney can be guaranteed. Lee et al. (100) introduced a nonlinear mixed-example augmentation method by fusing and mixing images from different classes to generate new training samples for standard plane classification of fetal US.

Fascinated by the superior performance of generative models in synthetic image generation, researchers have also implemented these models in the area of US image analysis (101, 102). To increase interpretability during image generation, Shi et al. (103) integrated expert annotations into a GAN to generate anatomy- and domain-specific synthetic US images for thyroid nodule classification. For a similar application, Zhang et al. (104) implemented a progressive GAN-based inpainting method to generate reliable structure and texture for thyroid US image analysis. Apart from GANs, variational autoencoders (VAEs) are another alternative for image synthesis. Pesteie et al. (105) employed a conditional VAE to augment data for classification of spinal US, where the label information and the image embedding are disentangled to increase the manipulability during synthesis. To realize deformable augmentation for 3D US volumes, Wulff et al. (106) implemented a conditional VAE to apply deformation on the 3D patch of US volumes. However, the development and use of augmentation methods, which do not explicitly take the physics of US into account, could result in unrealistic and sometimes physically wrong US images. The addition of such images could even reduce the performance and robustness of the US image processing. By explicitly taking the particular physics of US into account, Tirindelli et al. (107) introduced three physics-inspired augmentation strategies—deformation, reverberation, and signal-to-noise ratio—to generate plausible US images.

4.2. Physics-Inspired Machine Learning for Ultrasound

In the previous sections, we have shown how the latest breakthroughs in the field of machine learning and deep learning have elevated the conception of RUSSs to a higher level. Nonetheless, medical images are different from normal optical images. The intensity value of each pixel no longer represents only the color but also reflects the unique properties of human tissues. US, as a mechanical wave, has a much slower propagation speed than X-rays or light, which makes the resulting images very vulnerable to disturbances and full of noise. Such issues have posed extra challenges for a comprehensive understanding of US images. Thus, researchers have attempted to exploit extra clues from either the physiological context or US formation principles. Physiology-inspired approaches refer to network architectures in which prior clinical knowledge is explicitly integrated to facilitate a deep understanding of the US images. On the other hand, taking the US formation principles into consideration when designing the network structures appears to be another promising solution. By doing so, the US images are treated not as a distribution of grayscale values but as a representation of human tissues.

4.2.1. Physiology-inspired image analysis. Unlike normal optical images, US images are hard to interpret without sufficient medical knowledge. A deep understanding of anatomy is one of the key features that differentiates experienced sonographers from trainees. Inspired by the interpretation process of the medical experts, physiological knowledge is explicitly integrated into the deep learning–based US analysis network. To improve the segmentation performance of vessels in US images, Jiang et al. (25) proposed optical flow–based U-Net (OF-UNet), which explicitly utilizes the continuity of vessels. Guidance is provided to the network based on the optical flow image and the predicted segmentation mask from the previous frame. During breast US examination, sonographers utilize the information of previous frames to determine whether a lesion-like anatomy is indeed a diseased tissue or not. Based on such observations, Yu et al. (108) proposed UltraDet, which uses inverse optical flow progress to trace back in previous frames in order to detect negative symptoms that indicate false-positive classifications. To produce anatomically plausible semantic segmentation of cardiac US, Painchaud et al. (109) proposed a framework to guarantee the anatomical correctness of resulting segmentation maps. By transferring all the ground truth labels from image space to latent space using a conditioned VAE, a manifold of valid cardiac shapes

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in the feature space can be determined. Then, a mapping between the anatomically incorrect segmentation mask and the manifold of valid shapes is performed to improve the shape correctness. To correct the force-induced US image deformation, Jiang et al. (110, 111) explicitly considered tissue stiffness maps to compute an accurate and consistent anatomical geometry in RUSSs.

To resolve the challenge of detecting tiny tissues in US images (e.g., nerves), Dou et al. (112) proposed a pipeline that utilized anatomical knowledge to provide region proposals. Guided by the segmented surrounding organs (i.e., carotid artery, thyroid, and trachea), a Bayesian shape alignment module is applied to generate a region proposal for locating the recurrent laryngeal nerve. To generate temporal consistent segmentation results for cardiac US, Painchaud et al. (113) proposed a framework to perform postprocessing on the segmentation results. An autoencoder was introduced to map the segmentation masks into an interpretable latent space. For a sequence of cardiac US images, a smoothing process is applied to each attribute of the latent representations until a temporal consistency metric is satisfied. The periodic motions of arteries are frequently utilized by doctors as a strong prior for detection and localization in clinical practice. Huang et al. (17) employed such prior knowledge in the network design to facilitate the segmentation results for small arteries (i.e., radial arteries). Motion magnification techniques were introduced to extract a pulsation map, which highlights the region where the periodic motions occur. The pulsation map then serves as an attention map to guide the segmentation network.

4.2.2. Ultrasound physics-inspired image analysis. The grayscale values in US images represent the intensity of the reflected US waves. For visualization reasons, the source signals are mapped onto a color space between 0 and 255, so that they are interpretable by humans. A lot of information is lost during such a visualization process. However, this is not necessarily true for machines. With growing computational power, it is becoming feasible to learn how to interpret the source data directly. Therefore, Gare et al. (114) proposed W-Net, combining raw radio frequency data with B-mode images to optimize the segmentation results. By extracting information from the radio frequency data, W-Net is able to surpass the performance of a U-Net with a smaller network size. Another distinct property inherent to US is the Doppler effect. Doppler images are widely used in clinical practice to detect and measure blood flow. The Doppler effect has also been applied to provide weak supervision for the vessel segmentation network (115). Jiang et al. (62) later combined Doppler images with B-mode US images to provide extra guidance to the network for radial artery segmentation. Since the Doppler effect is unstable during autonomous US scanning, a RUSS was designed with the capability to reacquire a high-quality Doppler signal by performing out-of-plane fan motion.

A significant challenge in US imaging is the scarcity or absence of information in regions with attenuation or shadowing, a factor that substantially impedes the interpretability of these areas. Based on the principles of US wave propagation, Karamalis et al. (79) introduced confidence maps, which modeled the confidence assessment task as a random walk problem. The resulting confidence map was then utilized by several RUSSs as a criterion for US image quality optimization (32, 72, 116). Later, Klein & Wells (117) proposed to integrate radio frequency data into the random walk–based confidence estimation. Meng et al. (118) introduced a learning-based method to estimate the shadow areas for fetal US using a weakly supervised learning approach since it is not feasible to provide pixel-level shadow annotations from human operators.

Inspired by physics-based US simulation from CT segmentation maps, Velikova et al. (64) proposed to compose an intermediate CT–US space through US simulation parameterization. The segmentation is performed solely in the intermediate space, while CT and US images are transferred to such a common anatomical space through a specific parameterized US simulator during training and through a GAN during inference, respectively. Going one step further, instead

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of using hand-crafted parameterization for CT-to-US simulation, Velikova et al. (65) introduced a fully differentiable US simulator and realized an end-to-end training setting. The network was then able to optimize the intermediate space for different anatomy separately. Inspired by the neural radiance field (NeRF) (119), Wysocki et al. (120) presented Ultra-NeRF, which learned an intrinsic representation of each point in the US volume based on US physics, so that US images can be generated from new viewpoints.

4.3. Efficient Data Representation

The effectiveness of a CNN is dependent mainly on its ability to extract distinct features. The redundancy in feature space largely hinders the inference performance of deep learning models, especially when the diversity of the training dataset is limited. To increase the conciseness of the data representation, a disentanglement between task-related and task-irrelevant features is an essential step toward robust and reliable intelligent assessment of US images. Using V-Net (121) as the backbone, Degel et al. (122) proposed a 3D segmentation framework that uses a discriminator to predict the specific domain of the input US volume, and the corresponding adversarial loss forces the segmentation network to ignore the domain information. Meng et al. (123) also utilized adversarial loss to regularize the network to extract distinct features for corresponding classification tasks separately. The performance of the proposed segmentation network was validated on fetal US standard plane classification and shadow detection tasks. To increase the cross-domain robustness of the network, Meng et al. (124) implemented feature space clustering and demonstrated promising results in handling the domain shift in fetal US standard plane detection. Combining feature space alignment and an adversarial learning strategy, Ying et al. (125) proposed a classification scheme for benign and malignant thyroid nodules and showed high generalization ability across different US machines. For the same task, instead of using a distance-based feature alignment method, Zhang et al. (126) used a graph convolution network (127) to assess the relations between the source and target domain data and further realize cross-device generalization. Another, more intuitive way to decouple the domain- and task-related features is through minimizing a metric that can directly measure the shared information between the two variables. To this end, mutual information (128), which measures the dependency between two random variables, has been applied in various classification tasks in computer vision (129–131). In the US field, Meng et al. (132) realized unseen domain generalization for fetal US standard plane detection through mutual information minimization. Following a similar concept, Bi et al. (133) extended the usage of mutual information to segmentation networks by introducing a cross-reconstruction strategy.

5. OPEN CHALLENGES AND FUTURE PERSPECTIVES

Significant progress has been made toward achieving the ultimate goal of developing intelligent robotic sonographers by enhancing the understanding of the environment using advanced machine learning algorithms. However, some challenges remain for the research community, such as ethical and legal issues. Apart from those, emerging hardware design in the US area has revealed brand-new potential for future research. In this section, we discuss the ethical problems and provide insights for potential directions to stimulate future studies.

5.1. Ethics and Regulations

Given the rising enthusiasm within the market, the scientific community, and the medical field, ethical and regulatory guidance is becoming increasingly important. A clear definition of responsibility for such emerging systems could boost the research in the field of clinical translation,

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in order to bring the prototypes from the laboratories to the market. The demand for proper regulations has been discussed and emphasized by pioneers in the area of applied medical robotics (6, 134–138).

To tackle the ethical, regulatory, and legal issues for autonomous medical robots, Yang et al. (2) defined six levels of autonomy: no autonomy, robot assistance, task autonomy, conditional autonomy, high autonomy, and full autonomy. This taxonomy has been widely used to categorize the relevant techniques at their respective levels (136, 139). Different degrees of autonomy require distinct regulatory frameworks. In the case of medical robots with low autonomy levels, accountability rests primarily with the clinicians, who maintain complete control over the system. Due to this well-defined attribution of responsibility, the transition to commercialization for such systems tends to be more successful than it is for other systems. The da Vinci robot serves as the most prominent example of this success.

Concerning the regulation of medical systems that exhibit a certain level of autonomy, two leading standard-development organizations—the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC)—have published their inaugural joint standard (IEC/TR 60601-4-1) to address the safety, transparency, and trustworthiness of contemporary medical devices (136). Unlike surgical robots, robotic sonographers are noninvasive and interact with patients who are not under anesthesia; thus, more specialized regulations are necessary.

5.2. Emerging Ultrasound Imaging Systems

The fundamental developments in new types of US sensing hardware will significantly impact current clinical intervention procedures and bring new opportunities. In this section, we highlight the emerging optical US systems and soft ultrasonic patches.

5.2.1. Optical ultrasound. To enhance the versatility of US technologies, integrating optical systems with acoustic methods offers an intriguing path forward. By leveraging laser-induced thermal expansion within biological tissues, optoacoustic imaging enables high-resolution, molecule-specific imaging. This capability is crucial for distinguishing various tissue types, identifying pathological conditions, and pinpointing specific molecular markers (140). Building on recent progress, advancements in all-optical US (OpUS) transducers show particular promise for use in minimally invasive surgery. OpUS generates US via a fiber optic transducer using the photoacoustic effect and captures tissue reflections through optical interferometry, akin to optical coherence tomography (OCT) methods (141). These transducers excel in their remarkable miniaturization while preserving high-resolution imaging and mechanical flexibility, making them especially useful in endovascular applications like thrombus imaging (142).

Notably, this technology allows real-time feedback, creating opportunities for robotic monitoring in surgical procedures and image-based diagnostics. A research team at Johns Hopkins University was the first to demonstrate a robotic tracking system guided by optoacoustic signals (143, 144). The approach involves embedding an optical fiber into the distal end of a surgical instrument. This setup allows a robotic manipulator, equipped with a US probe, to monitor and trace the position of the tool tip through analysis of the optoacoustic signal. The authors applied this approach to robotic catheters (143) and needle tracking (144).

5.2.2. Ultrasonic patch. Given the recent advancements in unconventional US sensors such as US patches (145, 146), the field of US applications could be significantly expanded. These emerging soft US patch technologies, thanks to their size and flexibility, facilitate extended monitoring of patients. This allows for continuous capture of physiological data, aiding in accurate clinical

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diagnoses and close tracking of disease progression. Such extended monitoring was previously expensive and inefficient when utilizing traditional, bulky US probes.

In 2018, Hu et al. (145) presented a soft US patch consisting of a 10 × 10 array of piezoelectric transducers for 3D imaging of deep tissues (145). They reported that the patch has excellent electromechanical coupling, minimal cross-talk, and more than 50% stretchability. Wang et al. (146) applied a skin-conformal US patch array to monitor hemodynamic signals from tissues up to 14 cm beneath the skin. In another study, Wang et al. (147) presented a bioadhesive US array to provide 48 h of continuous imaging of various organs, including blood vessels, muscle, heart, diaphragm, lung, and so on. Focusing on cardiac imaging, Hu et al. (148) proposed a wearable US patch imager that is used to extract continuous signals for evaluating cardiac functions. When paired with a deep learning model for the automated segmentation of ventricular volume, this setup enables real-time calculations of essential cardiac performance metrics, including stroke volume, cardiac output, and ejection fraction. To provide wireless communication, Lin et al. (149) designed a miniaturized flexible control circuit that directly interfaces with the US patch. This allows continuous imaging of moving objects. These fundamental advancements in US sensors could unlock numerous opportunities for revolutionizing the techniques of robot-assisted US imaging.

6. CONCLUSION

This review has focused on highlighting the significance of machine learning and deep learning technologies in enhancing the intelligence of RUSSs. To provide an overview of the ongoing developments in the field, we initially discussed the specific robotic mechanisms and control systems employed in RUSSs, as well as their clinical applications. Modern robotic systems and control methods have already demonstrated superhuman levels of reproducibility. However, they still face challenges in understanding the physiological aspects of human tissue and diseases, different imaging modalities, imaging physics, and artifacts. Additionally, there is a need to improve their ability to promptly reason and make correct decisions in response to expected or unexpected changes during acquisition. To achieve this, we discussed two categories of methods for action reasoning, one based on implicitly interpreting input data and the other on explicitly interpreting it. Toward the conclusion of the review, we also discussed the open challenges and future perspectives.

The development and deployment of intelligent RUSSs will be a challenging and exciting endeavor. We believe robotic sonography and autonomous US examination represent the next frontier in medical robotics and hold significant promise in future clinical practices. Furthermore, we have highlighted the crucial importance of advancements in the area of ethics and regulations and also emphasized the need for collaborative efforts from the scientific, industrial, and clinical communities.

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